THE VALUATIVE CRITERION CHARACTERIZES UNIVERSALLY CLOSED QUASI-COMPACT QUASI-SEPARATED MORPHISMS OF ALGEBRAIC SPACES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-morphisms.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The author states that a formal argument gives the reduction to the case of X_A over Spec(A); the reduction is displayed as diagrams in the counted span and the paragraph that follows carries out the remaining work in full.

Editorial note. The four references of the counted spans are recorded as non-load-bearing navigation pointers; the argument that consumes them stays inside the counted proof span.

Editorial note. Preamble: the three native excerpt spans of sources/stacks-spaces-morphisms/preamble.tex (lines 8-13, 14-18 and 154-217). The chapter file itself carries no preamble text.

Allowed standard prerequisites (editorial).

- The converse criterion: quasi-compact plus valuative criterion implies universally closed. Lemma lemma-quasi-compact-existence-universally-closed, used at proof lines 8875-8877. Turns assertion (1) into assertion (2).
- Reaching points of the scheme theoretic image of a morphism from a valuation ring. Lemma lemma-reach-points-scheme-theoretic-image, used at proof lines 8914-8922. Produces the extension A' of A dominating A with the closed point mapping to x'.
- Permanence of quasi-compactness for the morphism x : Spec(K) -> X. Lemma lemma-quasi-compact-permanence, used at proof lines 8906-8908. Makes x quasi-compact, so that the closure of the image of x is the image of a closed set.
- Base change stability of quasi-separatedness and universal closedness. used at proof lines 8894-8895. Justifies the formal reduction to the case Y = Spec(A).
- The shape of the valuative criterion diagram. Definition definition-valuative-criterion, used at proof lines 8878-8893. Fixes the diagrams that have to be completed.

Source reference. Original definition 03IX: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4619–4619. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 03KA: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4670–4670. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Date: stacks-project:spaces-morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 03KS: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4688–4688. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 089B: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 10491–10491. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Lemma 0.2. *Let S be a scheme. Let $f : X \to Y$ be a morphism of algebraic spaces over S.*

(1) *If f is quasi-separated and universally closed, then f satisfies the existence part of the valuative criterion.*
(2) *If f is quasi-compact and quasi-separated, then f is universally closed if and only if the existence part of the valuative criterion holds.*

Proof. If (1) is true then combined with Lemma 03KA we obtain (2). Assume f is quasi-separated and universally closed. Assume given a diagram

$$\begin{array}{ccc} \mathrm{Spec}(K) & \longrightarrow & X \\ \downarrow & & \downarrow \\ \mathrm{Spec}(A) & \longrightarrow & Y \end{array}$$

as in Definition 03IX. A formal argument shows that the existence of the desired diagram

$$\begin{array}{ccccc} \mathrm{Spec}(K') & \longrightarrow & \mathrm{Spec}(K) & \longrightarrow & X \\ \downarrow & & & & \downarrow \\ \mathrm{Spec}(A') & \longrightarrow & \mathrm{Spec}(A) & \longrightarrow & Y \end{array}$$

follows from existence in the case of the morphism $X_A \to \mathrm{Spec}(A)$. Since being quasi-separated and universally closed are preserved by base change, the lemma follows from the result in the next paragraph.

Consider a solid diagram

$$\begin{array}{ccc} \mathrm{Spec}(K) & \xrightarrow{x} & X \\ \downarrow & \nearrow & \downarrow f \\ \mathrm{Spec}(A) & = & \mathrm{Spec}(A) \end{array}$$

where A is a valuation ring with field of fractions K. By Lemma 03KS and the fact that f is quasi-separated we have that the morphism x is quasi-compact. Since f is universally closed, we have in particular that $|f|(\overline{\{x\}})$ is closed in $\mathrm{Spec}(A)$. Since this image contains the generic point of $\mathrm{Spec}(A)$ there exists a point $x' \in |X|$ in the closure of x mapping to the closed point of $\mathrm{Spec}(A)$. By Lemma 089B we can find a commutative diagram

$$\begin{array}{ccc} \mathrm{Spec}(K') & \longrightarrow & \mathrm{Spec}(K) \\ \downarrow & & \downarrow \\ \mathrm{Spec}(A') & \longrightarrow & X \end{array}$$

such that the closed point of $\mathrm{Spec}(A')$ maps to $x' \in |X|$. It follows that $\mathrm{Spec}(A') \to \mathrm{Spec}(A)$ maps the closed point to the closed point, i.e., A' dominates A and this finishes the proof. $\square$